

Solid Organozinc Pivalates: A New Class of Zinc Organometallics with Greatly Enhanced Air- and Moisture-Stability

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Dedicated to Prof. Herbert Mayr on the occasion of his 70th birthday



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Abstract Organozinc species are powerful reagents for performing carbon–carbon and carbon–heteroatom bond-forming reactions in the presence of a transition-metal catalyst. However, extended applications of zinc reagents have been hampered by their moderate air- and moisture-stability. This short review presents our recent developments on the preparation of solid aryl, benzyl, heteroaryl, allyl zinc pivalates and zinc amide enolate reagents with greatly enhanced stability toward air and moisture.

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Key words zinc, magnesium, cross-coupling, transition metal, metalation

1 Introduction

Organozinc reagents were introduced into organic chemistry by Frankland¹ and they are intimately linked to the development of modern organic synthesis. The first organozinc compound reported by Frankland (Et_2Zn)² has excellent nucleophilic properties, reacting readily with diethyl oxalate,² but also proved difficult to handle safely. In fact, neat diethylzinc is highly air- and moisture-sensitive; it burns spontaneously in air and may react explosively with

oxygen.³ Secondary dialkylzincs are even photolabile and must be kept in the dark.⁴ However, long-chain dialkylzincs are easier to handle and are not spontaneously flammable under standard laboratory conditions.⁵ Furthermore, organozinc halides do not ignite spontaneously and can be handled without special precautions under argon. Nevertheless, these zinc organometallics are sensitive to moisture and air and display stabilities that are comparable to Grignard reagents. This sensitivity toward air and moisture has become a major synthetic drawback in recent years. It became clear that organozinc reagents undergo smooth transmetalations with various transition metals such as copper(I), nickel(II), palladium(II), and cobalt(II) salts.⁶ In fact, the Negishi cross-coupling (e.g., the palladium-catalyzed cross-coupling between unsaturated zinc reagents and unsaturated halides)⁷ constitutes one of the best methods for linking carbon–carbon bonds between unsaturated moieties, which is a key transformation for the preparation of numerous pharmaceuticals and agrochemicals. In recent years, a number of synthetic methods have become available for preparing polyfunctional organozinc halides, enhancing the scope of all carbon–carbon bond-forming reactions involving organozinc reagents.⁸ Therefore, there is a real need to prepare highly functionalized organozinc reagents with enhanced air- and moisture-stability. Furthermore, alternative cross-coupling reagents, for example boronic acids and esters, are more reluctant to undergo transmetalations with transition metals (they often require the presence of a base or a fluoride) and several of these boron derivatives have genotoxic properties.⁹ Some reports¹⁰ have shown clearly that organozinc derivatives bearing an oxygen–zinc bond should be less reactive toward air and moisture and this stimulated our research work on the search of air-stable organozinc reagents with oxygenated ligands attached to zinc.¹¹ In this short review, we describe our efforts towards the synthesis of organozinc pivalates, a new



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class of air- and moisture-stable reagents as well as their use for forming new carbon–carbon bonds in an efficient manner.

2 Preparation of Organozinc Pivalates

2.1 Using Organic Halides as Substrates

Aryl and heteroaryl iodides or bromides of type **1** are readily converted into the corresponding organozinc pivalates of type **2**^{12,13} by using either a direct insertion of mag-

nesium turnings¹⁴ in the presence of LiCl (method A; Figure 1), or a halogen/magnesium exchange using *i*PrMgCl·LiCl¹⁵ (method B). The best preparation of aryl- or heteroarylzinc pivalates by an oxidative addition method typically consists of treating an aryl bromide such as **1a** with magnesium turnings (2.5 equiv) and anhydrous LiCl (1.25 equiv) in THF at 25 °C for 2 hours. Addition of Zn(OPiv)₂ at 25 °C for 15 minutes produces the corresponding arylzinc pivalate (**2a**) in 92% yield, as indicated by iodometric titration (Scheme 1).

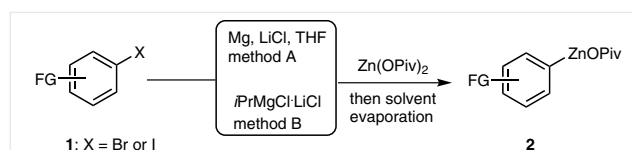
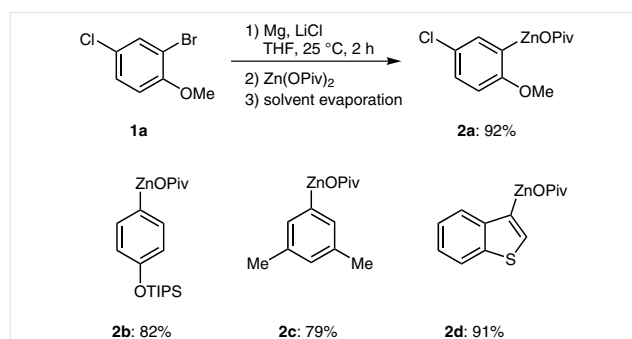
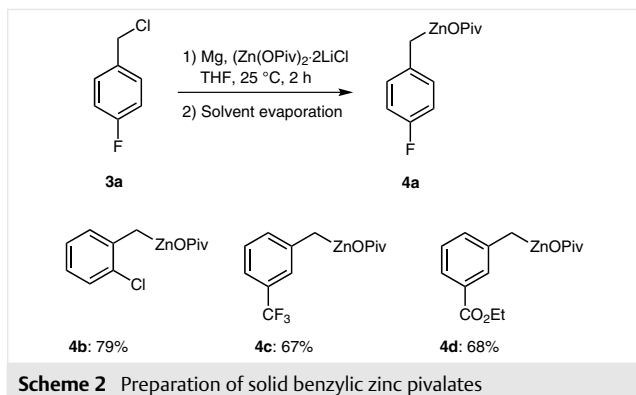


Figure 1 Synthesis of arylzinc pivalates through Mg insertion (method A) or Hal/Mg exchange (method B) followed by transmetalation with Zn(OPiv)₂

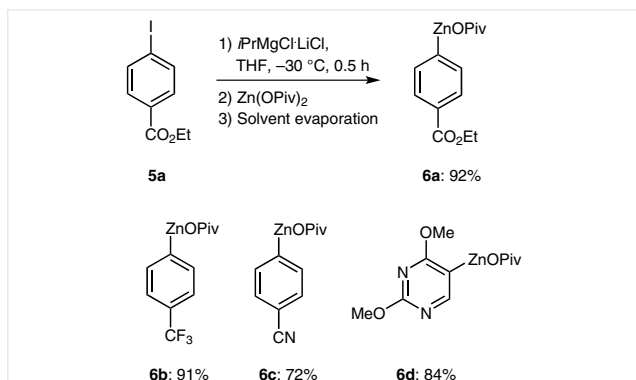


Scheme 1 Preparation of solid organozinc pivalates by Mg insertion

The aryl- and heteroarylzinc pivalates **2b–d** were prepared similarly. Structural studies performed by Mulvey and Hevia indicate that the exact structure of the zinc species is much more complex than RZnOPiv and that these reagents are better represented as RZnHal·Mg(OPiv)₂·*n*LiCl (Hal = Cl, Br or I); suggesting also that the initial motivation for preparing air-stable zinc reagents by attaching an oxygen ligand to zinc may only be partially justified and that the presence of Mg(OPiv)₂ may furthermore be responsible for the exceptional air-stability of these new organozinc reagents.¹⁶ Interestingly, the handling of such solid organozinc reagents in air for ca. 4 hours led to less than 10% decomposition. The method has been applied to the preparation of solid benzylic zinc pivalates. Thus, treatment of 4-fluorobenzyl chloride (**3a**) with magnesium turnings in the presence of Zn(OPiv)₂·2LiCl at 25 °C produces, after solvent evaporation, the desired benzylic zinc pivalate (**4a**) in 80% yield, as determined by iodometric titration.¹³ Similarly, the benzylic zinc pivalates **4b–d** were prepared in ca. 70–80% yield (Scheme 2).



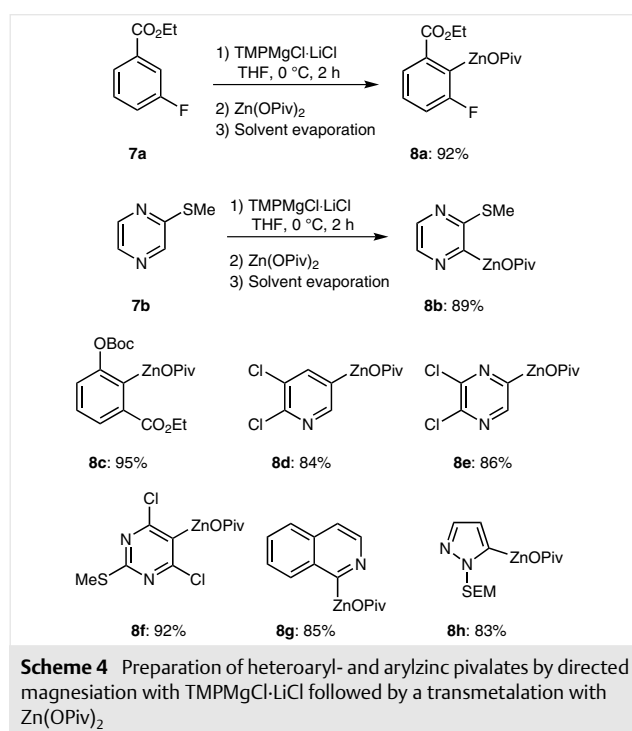
A second method using aryl bromides or iodides such as **5a** as substrates consists of performing a halogen/magnesium exchange followed by the addition of Zn(OPiv)_2 . In the case of aryl iodide **5a**, the I/Mg exchange is performed at $-30\text{ }^\circ\text{C}$ with $i\text{PrMgCl}\cdot\text{LiCl}$ ^{12,15} and is complete at this temperature within 30 minutes. Addition of Zn(OPiv)_2 provides the corresponding functionalized arylzinc pivalate **6a** in 92% yield. By using similar reaction conditions, aryl- and heteroarylzinc pivalates **6b–d** were prepared in 70–90% yield (Scheme 3).¹²



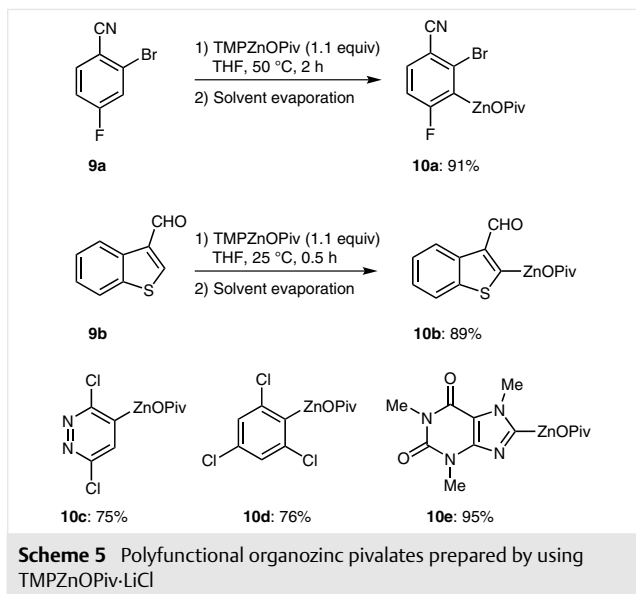
2.2 Using a Directed Metalation on Functionalized Arenes and Heteroarenes

The directed metalation of aromatic and heteroaromatic substrates¹⁷ is a highly expeditious way for converting a range of unsaturated molecules into the corresponding organozinc pivalates.¹⁸ A convenient procedure consists of treating an arene or heteroarene of type **7** with $\text{TMPMgCl}\cdot\text{LiCl}$ ¹⁹ (TMP = 2,2,6,6-tetramethylpiperidyl) in THF followed by the addition of Zn(OPiv)_2 and subsequent solvent evaporation. The resulting solid aryl- and heteroarylzinc pivalates of type **8** are produced in high yields.¹⁸ Thus, the treatment of ethyl 3-fluorobenzoate (**7a**) with $\text{TMPMgCl}\cdot\text{LiCl}$ (1.1 equiv) in THF at $0\text{ }^\circ\text{C}$ for 2 hours produces an intermediate magnesium reagent, which, after addi-

tion of Zn(OPiv)_2 (1.2 equiv, $25\text{ }^\circ\text{C}$, 15 min), furnishes the corresponding arylzinc pivalate **8a** in 92% yield, as indicated by iodometric titration. A heterocyclic substrate such as pyrazine (**7b**) is similarly fully magnesiated with $\text{TMPMgCl}\cdot\text{LiCl}$ at $-20\text{ }^\circ\text{C}$ for 4 hours and gives, after transmetalation with Zn(OPiv)_2 and solvent evaporation, the solid heteroarylzinc pivalate **8b** in 89% yield (Scheme 4). The presence of 2,2,6,6-tetramethylpiperidine in the solid organozinc pivalates does not affect their air- and moisture-stabilities. By using the same metalation procedure, it was possible to prepare a wide range of organozinc pivalates such as **8c–h** (Scheme 4).^{12,18}



A directed zincation was achieved by using the TMP-base $\text{TMPZnCl}\cdot\text{Mg(OPiv)}_2\cdot\text{LiCl}$ conveniently abbreviated as $\text{TMPZnOPiv}\cdot\text{LiCl}$. This zinc amide base is quite active and well soluble in THF (0.85 M)²⁰ compared with $\text{TMPZnCl}\cdot\text{LiCl}$ (ca. 1.4 M).^{21,22} It is readily prepared by treating $\text{TMPMgCl}\cdot\text{LiCl}$ with Zn(OPiv)_2 in THF at $0\text{ }^\circ\text{C}$ (0.5 h). Unfortunately, this zinc base is not stable as a solid for a long time, but in 0.85 M solution in THF it is quite stable at $0\text{ }^\circ\text{C}$ for several days. Treatment of functionalized benzonitrile **9a** with $\text{TMPZnOPiv}\cdot\text{LiCl}$ at $50\text{ }^\circ\text{C}$ for 2 hours produces, after solvent evaporation, solid arylzinc pivalate **10a** in 91% yield.^{12,20} Sensitive functional groups such as an aldehyde function are well tolerated, and 3-formylbenzothioephene (**9b**) is readily zincated with $\text{TMPZnOPiv}\cdot\text{LiCl}$ at $25\text{ }^\circ\text{C}$ within 30 minutes, furnishing, after solvent evaporation, heterocyclic zinc pivalate **10b** in 89% yield (Scheme 5).

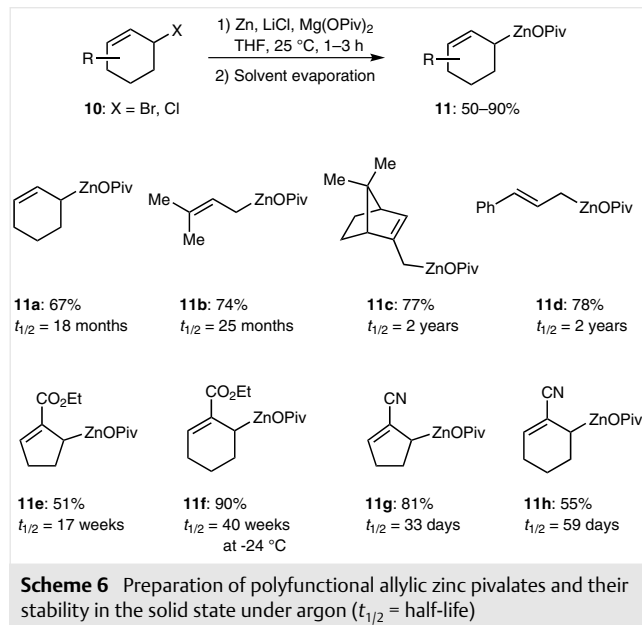


How unique is the pivalate anion for ensuring air-stability of the corresponding solid organozinc pivalates? To answer this question, we have prepared several organozinc derivatives (RZnY) in which Y was either a carboxylate or an amido derivative and have found that an excess of LiCl diminishes their air- and moisture-stability. However, the use of an amido derivative ($Y = N(R^1)COR^2$) has a positive effect on this stability and an improved compatibility with moisture and air was found when an amido group such as $Y = N[(CH_2)_2OMe]CotBu$ was used.¹⁸ The presence of further additional ligands such as dioxane or 1,2-dimethoxyethane also improves the air stability for various 2-pyridylzinc pivalates, which proved to be valuable reagents for performing palladium-catalyzed cross-couplings.²³

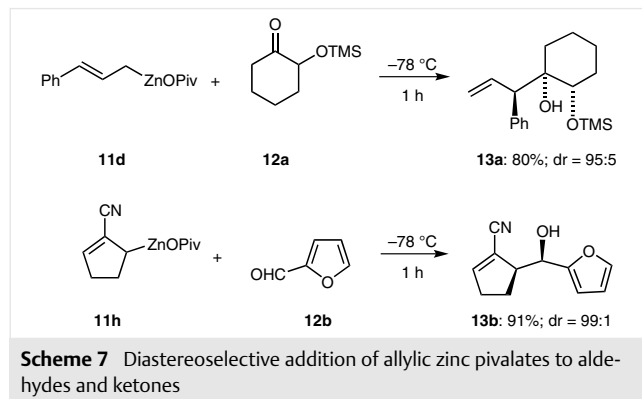
2.3 Preparation of Solid Allylic Zinc Pivalates

Allylic organometallics always exhibit a higher reactivity than the corresponding alkyl, aryl or even benzylic organometallic reagents due to a better stabilization of the negative charge on the allylic moiety, which leads to a more polar and more reactive carbon–zinc bond. Allylic zinc reagents have found numerous applications in organic synthesis.^{24,25} The preparation of solid allylic zinc pivalates is therefore of great importance. It was found that the treatment of various allylic chlorides or bromides with zinc powder in the presence of lithium chloride and magnesium pivalate ($Mg(OPiv)_2$) in THF provides the corresponding allylic zinc pivalates in 50–90% yield within 1 to 3 hours reaction time at room temperature.²⁶ After solvent evaporation, white to yellow powders of the solid zinc reagents of type **11** are obtained which are not stable in air, but can be stored for a long time (several months to years) at room

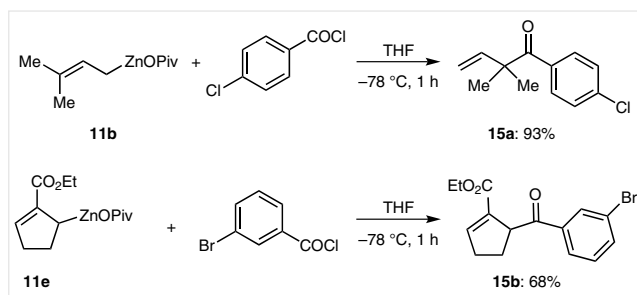
temperature or better at -24°C . Various substitution pattern of the double bond and of the ring system, including the presence of functional groups on the allylic double bond are possible (see **11a–d**), and the method provides quite general access to polyfunctional allylic zinc reagents such as **11e–h** (Scheme 6).



The allylic zinc reagents **11e–h**, bearing a functional group such as an ester or a nitrile, are significantly less stable and should be stored in a freezer (-24°C). Compared with regular organozinc pivalates, these allylic zinc derivatives display a significantly higher reactivity and they readily add to aldehydes or ketones. Thus, treatment of cinnamylzinc pivalate (**11d**) with the racemic ketone **12a** produces the tertiary alcohol **13a** bearing three contiguous stereocenters in 80% yield with excellent diastereoselectivity. Similarly, the functionalized zinc pivalate **11h** reacts with furfural (**12b**) to produce the unsaturated nitrile **13b** diastereoselectively in 91% yield ($dr > 99:1$; Scheme 7).²⁶



As usual, the new carbon–carbon bond is formed from the sterically more substituted end of the allylic system and only one regioisomer is obtained (see **13a**). Similarly, these allylic zinc pivalates react regioselectively with acid chlorides at $-78\text{ }^{\circ}\text{C}$. Under these conditions, no double addition to the carbonyl group is observed. Thus, prenylzinc pivalate (**11b**) adds to benzoyl chloride (**14a**) producing ketone **15a** regioselectively in 93% yield (Scheme 8).²⁶ Functionalized allylic zinc pivalates react in the same way, and the reaction of **11e** with acid chloride **14b** furnishes the corresponding polyfunctional ketone **15b** in 68% yield (Scheme 8).



Scheme 8 Regioselective acylation of allylic zinc pivalates

3 General Reactivity Patterns of Organozinc Pivalates

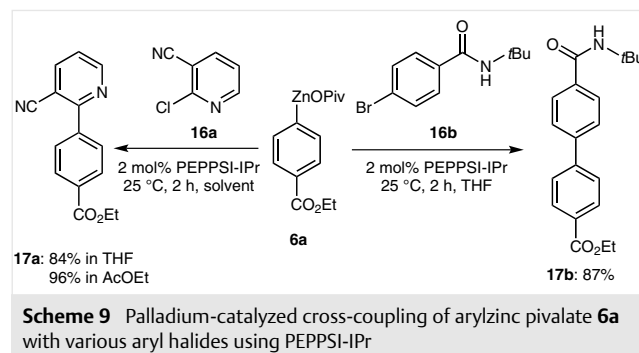
3.1 General Aspects

Solid aryl- and heteroarylzinc pivalates are excellent reagents for forming new carbon–carbon bonds and they display a similar reactivity to organozinc halides and diorganozincs. Only in a few cases, such as acylation reactions, does the presence of the pivalate anion interfere with the desired reaction pathway. Generally speaking, organozinc pivalates are much more robust zinc organometallic reagents and numerous reactions can be conducted under preparatively simpler procedures (for example, the water content of the solvents is less important for the outcome of the reaction). Organozinc pivalates have proven to be especially useful for the performance of transition-metal-catalyzed cross-couplings. The use of these zinc reagents for acylation, 1,4-addition and related carbon–carbon forming bonds will also be presented.

3.2 Transition-Metal-Catalyzed Cross-Couplings

Transition-metal-catalyzed cross-couplings are among the best methods for linking two unsaturated units, and numerous variations of this reaction have been described.²⁷ Palladium-catalyzed cross-couplings have found the broadest applications because of the reliability of such catalysts and availability of ligand families, which permit cross-couplings in difficult cases. Pioneering work of Farina,²⁸

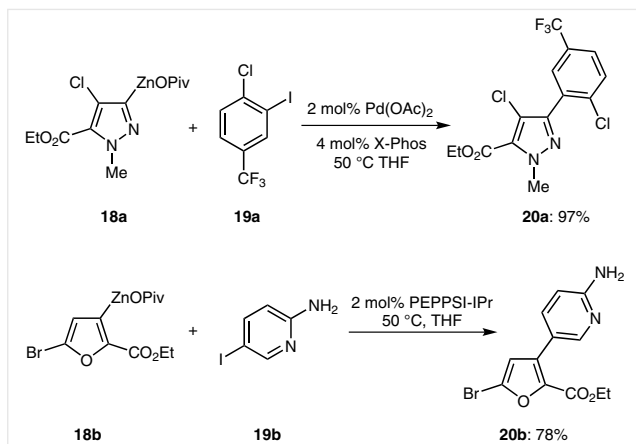
Hayashi,²⁹ Buchwald,³⁰ Organ,³¹ and others³² have made Suzuki,³³ Negishi,³⁴ Kumada,³⁵ Hiyama,³⁶ and Denmark³⁷ cross-couplings very useful reactions with numerous industrial applications especially in the pharmaceutical and agrochemical world.³⁸ Although boronic esters and related reagents have been mostly used, their variable stability, geno-toxicity⁹ and difficult synthesis has led to the search for alternative organometallic reagents, and organozinc pivalates proved to be an excellent alternative. Thus, treatment of arylzinc pivalate **6a** (see Scheme 3) with aryl and heteroaryl chlorides and bromides **16a,b** in the presence of the palladium complex developed by Organ (2% PEPPSI-IPr) at $25\text{ }^{\circ}\text{C}$ furnishes the desired cross-coupling products **17a,b** within 2 hours reaction time. Notably, the presence of the free amide function for the electrophile **16b** is perfectly tolerated in the cross-coupling and there is no need to use an excess of organozinc pivalate **6a**. Interestingly, this cross-coupling can be performed in both THF and ethyl acetate. In fact, ethyl acetate proved to be an excellent solvent for such palladium-catalyzed cross-couplings. The availability of **6a** as a solid powder greatly facilitates the solvent optimization for such cross-couplings (Scheme 9).¹³



Scheme 9 Palladium-catalyzed cross-coupling of arylzinc pivalate **6a** with various aryl halides using PEPPSI-IPr

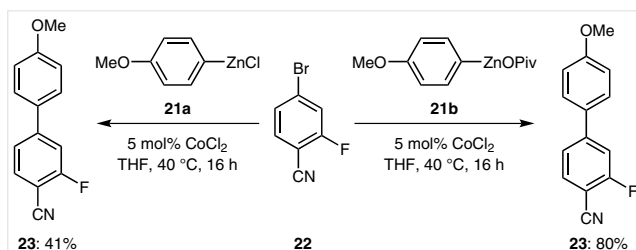
Polyfunctional coupling products are readily obtained because various highly functionalized aryl- and heteroarylzinc pivalates are available. Thus, pyrazolylyl zinc pivalate **18a** undergoes smooth cross-coupling¹⁸ with aryl iodide **19a** at $50\text{ }^{\circ}\text{C}$ using the ligand X-Phos developed by Buchwald.³⁸ Similarly, polyfunctional furylyl zinc pivalate **18b** undergoes a chemoselective cross-coupling with aminopyridine **19b** to provide the desired cross-coupling product **20b** in 78% yield (Scheme 10).¹⁸

Although palladium catalysis of organozinc pivalates is a carbon–carbon bond-forming reaction with broad applicability, the alternative use of CoCl_2 ^{39,40} as catalyst is quite advantageous because the price ratio $\text{PdCl}_2/\text{CoCl}_2$ is ca. 1000. The presence of a carbonyl ligand⁴¹ accelerates cobalt-catalyzed cross-couplings and enhances the scope of the reaction. Thus, comparing the cross-couplings of *p*-anisylzinc chloride (**21a**) or *p*-anisylzinc pivalate (**21b**) with aryl bromide (**22**) shows that the organozinc pivalate (**21b**) leads to a twofold better yield (Scheme 11).⁴² This cobalt-catalyzed

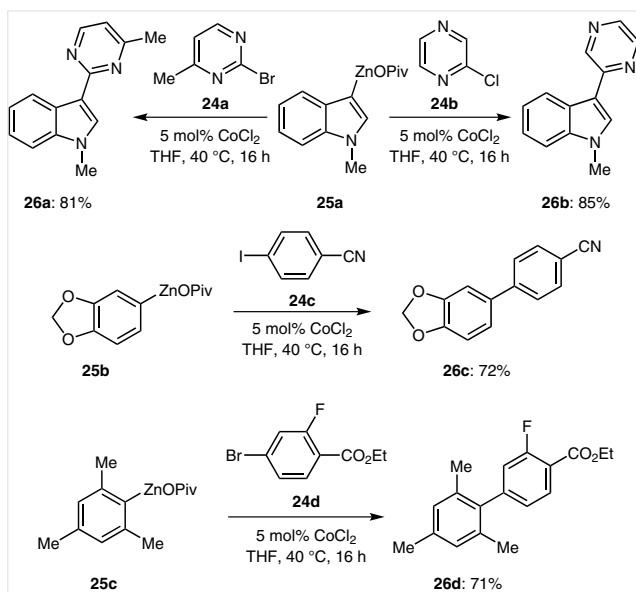


Scheme 10 Palladium-catalyzed cross-coupling of heterocyclic zinc pivalates

cross-coupling is completed within 16 hours at 40 °C and produces the desired cross-coupling product **23** in 80% isolated yield.



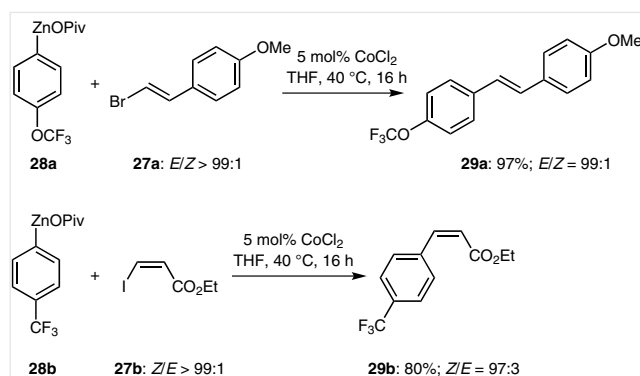
Scheme 11 Comparative cobalt-catalyzed cross-coupling using either an arylzinc chloride or the corresponding organozinc pivalate



Scheme 12 Cobalt-catalyzed cross-coupling of aryl- or heteroarylzinc pivalates with unsaturated halides

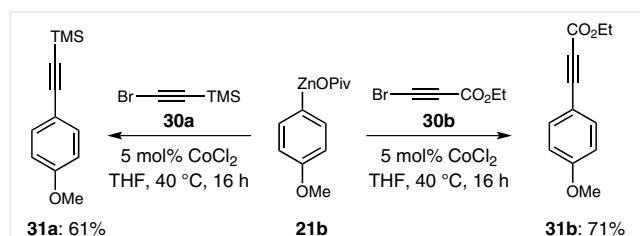
It turns out that this cobalt-catalyzed cross-coupling has a quite good reaction scope. Whereas electron-deficient *N*-heterocyclic halides such as **24a,b** undergo smooth cross-coupling with 3-indolylzinc pivalate (**25a**) to produce the expected coupling products **26a,b** in 81–85% yield (Scheme 12), various electron-deficient aryl iodides or bromides such as **24c,d** react well with organozinc pivalates **25b,c** under the same conditions, furnishing the cross-coupling derivatives **26c,d** in 71–72% yield (Scheme 12).⁴²

The cross-coupling reaction can be extended to alkenyl iodides or bromides. Thus, the cobalt-catalyzed cross-coupling of *E*-bromostyrene **27a** with arylzinc pivalate **28a** furnishes *E*-stilbene (**29a**) in 97% yield and *E/Z* = 99:1. The *Z*-iodoacrylate **27b** undergoes a similar cross-coupling with the CF₃-substituted arylzinc pivalate **28b**, affording *Z*-acrylate **29b** in 80% yield and *E/Z* = 3:97 (Scheme 13).⁴²



Scheme 13 Stereoselective cobalt-catalyzed cross-coupling of alkenyl halides with arylzinc pivalates

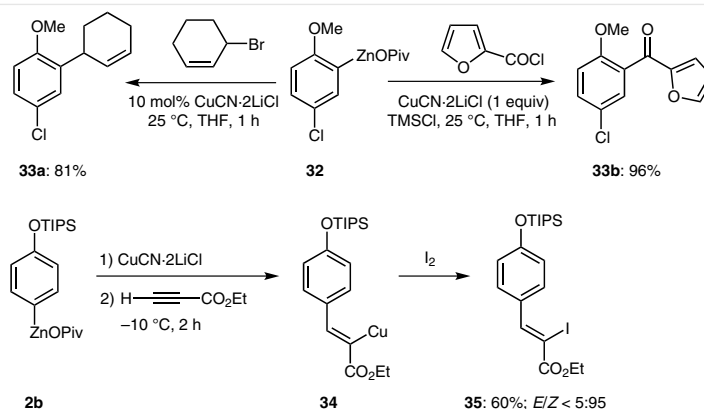
Finally, the highly reactive alkynyl bromides **30a,b** undergo smooth cross-coupling at –40 °C with *p*-anisylzinc pivalate (**21b**) providing arylacetylene derivatives **31a,b** in 61–71% yield (Scheme 14).⁴²



Scheme 14 Cobalt-catalyzed cross-coupling of alkynyl bromides and an arylzinc pivalate

3.3 Other Carbon–Carbon Bond-Forming Reactions Using Organozinc Pivalates

As for organozinc halides, organozinc pivalates readily undergo transmetalations with copper cyanide complexed with lithium chloride (CuCN·2LiCl).⁴³ Thus, arylzinc pivalate **32** is allylated by 3-bromocyclohexane in the presence of

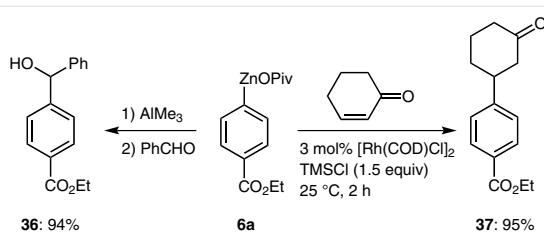


Scheme 15 Copper(I)-mediated reactions of arylzinc pivalates

10% CuCN·2LiCl to provide the allylated product **33a** in 81% yield.¹² Arylzinc pivalates are also readily acylated by various acid chlorides. In this case, an excess of TMSCl (6 equiv) is required to convert the arylzinc pivalate into the corresponding arylzinc chloride and PivOSiMe₃ in situ. Thus, treatment of **32** with furfuryl chloride in the presence of TMSCl and CuCN·2LiCl (1.0 equiv) provides the corresponding ketone **33b** in 96% yield (Scheme 15).

The carbometallation of activated alkynes⁴⁴ can be achieved with copper reagents derived from organozinc pivalates. Thus, the transmetalation of arylzinc pivalate **2b** with CuCN·2LiCl followed by the addition of ethyl propiolate at –10 °C in THF produces, after 2 hours reaction time, the copper intermediate **34**, which, after iodolysis, affords polyfunctional cinnamate **35** in 60% yield (*Z/E* > 95:5).¹²

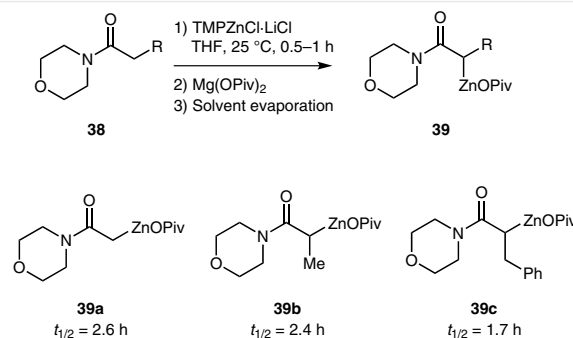
Organozinc pivalates react only sluggishly with aldehydes. However, treatment with trimethylaluminum, according to Woodward,⁴⁵ in a toluene/THF mixture followed by the addition of benzaldehyde furnishes the corresponding benzylic alcohol in satisfactory yield. Thus, treatment of polyfunctional arylzinc pivalate **6a** with AlMe₃, followed by addition of benzaldehyde, provides alcohol **36** in 94% yield. The same arylzinc pivalate **6a** undergoes a rhodium(I)-catalyzed 1,4-addition to cyclohexanone according to the method of Frost,⁴⁶ and Hayashi and Miyaura⁴⁷ (Scheme 16).¹³



Scheme 16 Further transformations of arylzinc pivalates

3.4 Preparation and Reactions of Solid, Salt-Stabilized Zinc Amide Enolates as New, Convenient Reformatsky Reagents

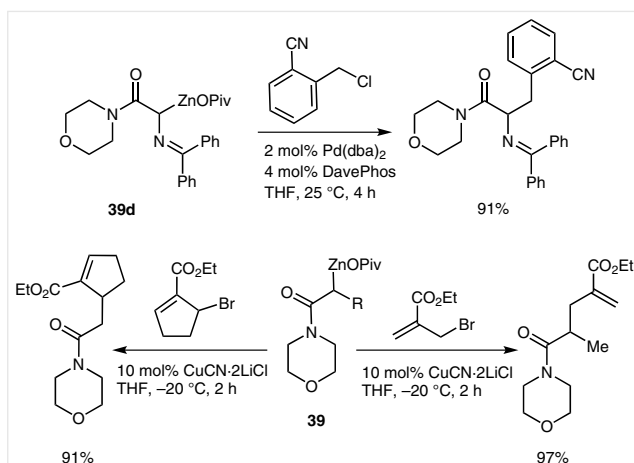
Reformatsky reagents⁴⁸ have found numerous applications in organic synthesis. Performing the deprotonation of *N*-morpholino amides of type **38** with TMPZnCl·LiCl produces the corresponding pivaloyloxy zinc enolates **39**, which are stable at room temperature in strong contrast to regular Reformatsky enolates, which are best used directly after their preparation⁴⁹ (Scheme 17).



Scheme 17 Preparation of pivaloyloxy zinc *N*-morpholino amides and their half-lives in air at 25 °C

A variety of zincated substituted *N*-morpholino amides (**39a–c**) can be prepared and have excellent air stabilities (up to 2 h in air at 25 °C as a solid; Scheme 17). They undergo palladium-catalyzed arylations and benzoylations as well as copper-catalyzed allylations (Scheme 18).

Finally, these new zinc enolates have proved to be very useful for the preparation of a potent anti-breast cancer candidate **40** by using an addition of **39a** to a benzonitrile followed by a Nenitzescu reaction.⁵⁰ The *N*-morpholino amide **41** was converted into an aldehyde function by using CpZr(H)Cl (Scheme 19).



Scheme 18 Palladium- and copper-catalyzed cross-couplings using zincated *N*-morpholino amides of type **39**

4 Conclusion

We have shown that various aryl, heteroaryl, benzylic, and allylic zinc pivalates can be prepared as solid powders (Figure 1) by using several complementary preparation procedures involving organic halides or heterocycles and arenes as starting materials. A broad range of functional groups are tolerated in this new class of organozinc reagents. They react with a broad range of aryl or heteroaryl halides in the presence of palladium or cobalt catalysts. Copper-mediated acylations or carbometalations further extend their reactivity scope. It is expected that further

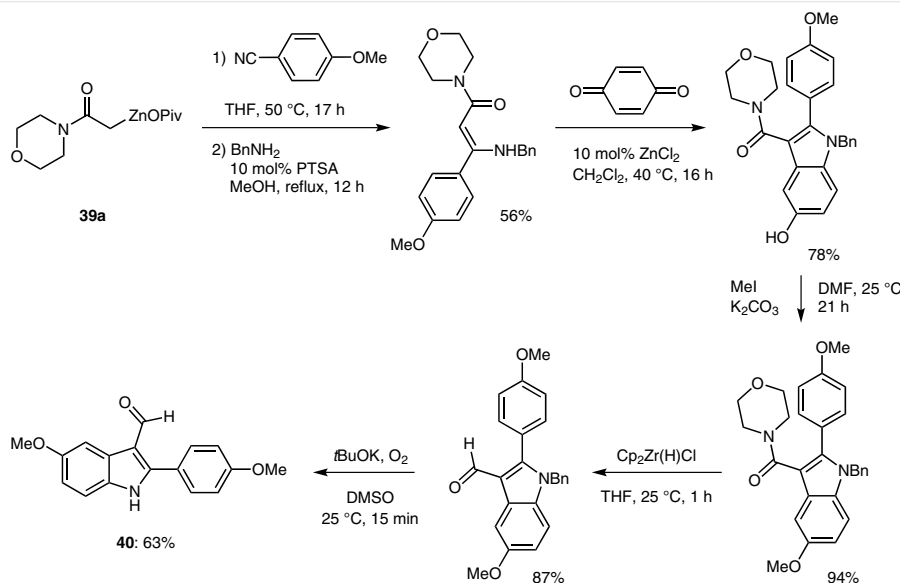
transformations will be developed in the future. Although, these organozinc pivalates have already found applications in high-throughput screening experiments involving palladium-catalyzed cross-couplings, further applications in pharmaceutical, agrochemical, and material chemistry are expected.

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Scheme 19 Synthesis of pharmaceutically relevant indole **40** by using zincated enolate **39a**

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